



UNIVERSIDAD
DE GRANADA

Final activity: Action Against Hunger

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Summary

In 2014, Guatemala's "Corredor Seco" faced a major humanitarian crisis caused by droughts and plagues, leading to critical levels of stunting.

This study employs data from the Guatemala SMART Survey and applies multivariate statistical techniques to characterize vulnerability in this real-world scenario.

The survey primarily focused on the nutritional status of children aged 6 to 59 months, a critical stage for physical and cognitive development. Consequently, this study evaluates which variables explain best how droughts and plagues affected the nutrition, food security, and mortality of children and their families.

Using dimension reduction techniques, PCA and FA, which explained 75% of the variance, variables related to biological growth, geographic location, and nutritional status were grouped into components. Subsequently, Cluster analysis was used to segment households. Validation of these profiles using Discriminant Analysis (DA) demonstrates that while geographic location and economy define the structural context, the effective identification of stunting depends almost exclusively on biometric variables.

It is concluded that detection strategies must prioritize rigorous anthropometric monitoring, as these indicators offer significantly superior predictive capacity compared to socioeconomic factors in contexts of generalized vulnerability.

Introduction

The "Corredor Seco" region in Guatemala represents one of the scenarios of greatest climatic and social vulnerability in Central America. In 2014, this area faced a complex crisis marked by a prolonged 45-day drought and the coffee leaf rust (*La Roya*) plague. These events exhausted basic grain reserves and drastically reduced the demand for agricultural labor, translating into profound financial uncertainty for the majority of rural households. This situation threatened to spike child mortality and malnutrition rates in communities that already had limited access to basic resources and healthcare services.

Although stunting remained at low severity levels (1.7% in the West and 3.1% in the East), GAM revealed a national emergency. In the western stratum, 62.3% of children under five suffered from GAM, evidencing a structural nutrient deficiency. The socioeconomic context exacerbates this reality: the near-exclusive dependence on agriculture forced many families to adopt emergency coping strategies, such as reducing the number of daily meals.

To evaluate this crisis, a cross-sectional study was conducted following the standardized *SMART Survey* methodology. The resulting database integrates biometric variables, morbidity indicators, and food security metrics, such as the **ELCSA** scale and the Coping Strategy Index (**CSI**). Given that the study of isolated variables is insufficient to understand complex phenomena, multivariate techniques were employed: PCA and FA to reduce dimensionality, cluster analysis to segment the population, and Discriminant Analysis (DA) to validate the identified profiles.

Within this framework, the present study aims to analyze the network of socio-biological factors that determine malnutrition in the region. The primary objective is to identify the latent dimensions of nutritional status and segment the population into risk profiles. This analysis seeks to contrast the predictive capacity of geographic location and economic situation against anthropometric indicators, evaluating whether these external factors act as critical differentiators of biological development or if nutritional vulnerability manifests transversally within the emergency context of the Corredor Seco.

Materials and Methods

0.1 Materials

The database used in this study is the result of merging two originally provided datasets. Following a process of coding, cleaning, and filtering, the dimensions of the dataset were adjusted, retaining only the variables of interest relevant to the stated objectives.

The final dataset includes information for 704 minors. The stored variables are categorized into the following groups:

- **Individual Information:** Each minor is associated with a unique identifier (ID_UNICO) and information regarding which of the two examined strata they belong to.
- **Biometric and Anthropometric Data:** This category includes age in months (MONTHS), weight in kilograms (WEIGHT), an analysis of weight relative to the mean (UNDERWEIGHT), height in centimeters (HEIGHT), sex, and mid-upper arm circumference (MUAC) in mm as a mortality risk indicator.
- **Health and Mortality Indicators:** Presence of conditions such as diarrhea, fever, bilateral edema, or respiratory infections; records of vaccinations and deworming; and the risk of stunting (SUNTING) and (GAM). Additionally, information regarding the child's dietary diversity is included (DIET_MIN_ACEP).
- **Household and Socioeconomic Variables:** Household size (TAMANHO_HOG) and food security indicators based on responses to the Household Food Insecurity Access Scale (stored in variables HOGAR_, ADULTOS_, MENORES_, ECSA_, and CSI_). This group also features family income analysis (NUM_FUENT_INGRES, EMPLEO_JORNAL_A1, EMPLEO_OTRO_B1, COMERCIO_C1, VENTA_PROD_D1) and the number of weeks of basic grain reserves (MAIZ_A and FRIJOL_B).

Prior to conducting the univariate study, we assessed the integrity of the database. For variables with a proportion of missing data exceeding 5%, a homogeneity study was performed by comparing the group with present data against the group with missing values.

Upon confirming a predominantly random pattern ($p > 0.05$), we proceeded with the imputation of these values to avoid reducing the statistical power of the sample. However, we are mindful of the limitations of this approach in variables where strict homogeneity was not fully met.

The following table provides a brief descriptive statistical analysis of the variables:

Variable <chr>	Mean <dbl>	Standard_deviation <dbl>
ESTRATO.x	1.599432	0.4903620
CLUSTER.x	15.110795	9.0400844
MONTHS	29.596776	17.3782306
WEIGHT	10.976034	3.4770672
HEIGHT	81.130813	13.5790171
MUAC	151.031250	13.3341620
TAMANHO_HOG	7.130868	3.4005949
MAIZ_A	6.944365	11.2961960
FRIJOL_B	3.557775	8.0665898
ELCSA_ADULT_CLASF	2.516035	0.8554878
ELCSA_MENS_CLASF	2.548387	0.8912227
NUM_FUENT_INGRES	1.148359	0.4045643
CSir2_OCC	2.569187	1.1109162
CSir2_ORI	2.563481	1.1061033
CSI_global	5.110795	2.2349092

Figure 1: Quantitative variables

Variable	Categoria	Freq	% Total	Variable	Categoria	Freq	% Total
SEX	f	352	50.00	ADULTOS_E	0	534	75.85
	m	349	49.57		1	167	23.72
	<NA>	3	0.43		<NA>	3	0.43
	Total	704	100.00		Total	704	100.00
EDEMA	n	697	99.01	ADULTOS_F	0	402	57.10
	y	3	0.43		1	297	42.19
	<NA>	4	0.57		<NA>	5	0.71
	Total	704	100.00		Total	704	100.00
DIARREA	0	500	71.02	ADULTOS_G	0	514	73.01
	1	200	28.41		1	185	26.28
	<NA>	4	0.57		<NA>	5	0.71
	Total	704	100.00		Total	704	100.00
FIEBRE	0	387	54.97	ADULTOS_H	0	583	82.81
	1	314	44.60		1	118	16.76
	<NA>	3	0.43		<NA>	3	0.43
	Total	704	100.00		Total	704	100.00
INFEC_RE	0	289	41.05	MENORES_I	0	321	45.60
	1	412	58.52		1	368	52.27
	<NA>	3	0.43		<NA>	15	2.13
	Total	704	100.00		Total	704	100.00
DESPARAS	0	361	51.28	MENORES_J	0	312	44.32
	1	247	35.09		1	376	53.41
	<NA>	96	13.64		<NA>	16	2.27
	Total	704	100.00		Total	704	100.00
VACC_SAR_AC	0	151	21.45	MENORES_K	0	558	79.26
	1	553	78.55		1	142	20.17
	<NA>	0	0.00		<NA>	4	0.57
	Total	704	100.00		Total	704	100.00
SUNTING	0	309	43.89	MENORES_L	0	455	64.63
	1	380	53.98		1	243	34.52
	<NA>	15	2.13		<NA>	6	0.85
	Total	704	100.00		Total	704	100.00
GAM	0	680	96.59	MENORES_M	0	485	68.89
	1	9	1.28		1	213	30.26
	<NA>	15	2.13		<NA>	6	0.85
	Total	704	100.00		Total	704	100.00
UNDERWEIGHT	0	560	79.55	MENORES_N	0	566	80.40
	1	131	18.61		1	132	18.75
	<NA>	13	1.85		<NA>	6	0.85
	Total	704	100.00		Total	704	100.00
INTRO ALIM	0	70	9.94	MENORES_O	0	613	87.07
	1	634	90.06		1	87	12.36
	<NA>	0	0.00		<NA>	4	0.57
	Total	704	100.00		Total	704	100.00
DIET_MIN_ACE	0	661	93.89	EMPLEO_JORN	0	211	29.97
	1	43	6.11		1	490	69.60
	<NA>	0	0.00		<NA>	3	0.43
	Total	704	100.00		Total	704	100.00
HOGAR_A.y	0	96	13.64	EMPLEO_OTRO	0	512	72.73
	1	605	85.94		1	189	26.85
	<NA>	3	0.43		<NA>	3	0.43
	Total	704	100.00		Total	704	100.00
HOGAR_B	0	453	64.35	COMERCIO_C1	0	641	91.05
	1	248	35.23		1	60	8.52
	<NA>	3	0.43		<NA>	3	0.43
	Total	704	100.00		Total	704	100.00
HOGAR_C	0	311	44.18	VENTA_PROD_	0	635	90.20
	1	377	53.55		1	66	9.38
	<NA>	16	2.27		<NA>	3	0.43
	Total	704	100.00		Total	704	100.00
ADULTOS_D	0	301	42.76				
	1	388	55.11				
	<NA>	15	2.13				
	Total	704	100.00				

Figure 2: Categorical variables

0.2 Statistical methods

To analyze the data and fulfill the proposed objectives, the following multivariate statistical techniques have been applied. The entire development was carried out using the R programming language, implementing the following methods:

Firstly, a **Principal Component Analysis (PCA)** was performed. As a preliminary step to techniques such as clustering, this analysis generates as many Principal Components as there are (numerical) variables used. This is achieved through linear combinations of the original variables that maximize variance and remain orthogonal to one another. In R, *PCA\$rotation* represents the loadings (the coordinates of the variables with respect to the principal components).

PCA identifies the directions where the explained variance is highest. Consequently, it was decided to perform the PCA—along with the subsequent technique—on two different sets of variables to compare the results. Once the components are obtained both numerically and graphically, the first few components are selected (applying the *Rule of Abdi et al.*).

Subsequently, **Factor Analysis (FA)** was performed on the same two groups previously studied. Similar to PCA, the goal was to reduce dimensionality; however, in this case, the focus was on isolating unobservable common factors underlying the measured variables. Specifically, the minimum residual (*minres*) method was employed alongside a *varimax* orthogonal rotation.

A simplified structure was compared against a more complex one to determine which configuration minimized the uniqueness of the variables while maximizing the explained common variance. This process facilitated a more precise identification of latent variables (biological, nutritional, and geographical), ensuring that the final set of variables selected was the most representative of the reality of malnutrition in the region.

To classify the population, a **Cluster Analysis (CA)** was implemented. *Hierarchical analysis* was used for sample segmentation, allowing the identification of groups of households or individuals with similar vulnerability profiles and coping strategies. The objective was to segment the sample into homogeneous groups to identify risk typologies based on health variables and socioeconomic factors.

This approach reveals patterns of vulnerability and coping strategies that differentiate high-morbidity groups from those with greater financial stability, thereby overcoming the limitations of traditional descriptive statistics.

Lastly, a **Discriminant Analysis (DA)** was conducted to summarize the sample, alongside Shapiro-Wilk tests and Q-Q plots to assess multivariate normality with respect to a selected categorical variable. Linear Discriminant Analysis (LDA) and Quadratic Discriminant Analysis (QDA) models were implemented, with the selection between them based on the homogeneity of covariance matrices.

Validation was performed by splitting the data into training (80%) and test (20%) sets. Model performance was evaluated using confusion matrices, the Apparent Error Rate (APER), and partition maps for both biometric and socioeconomic variables.

Results

0.1 Principal Component Analysis

To compare the results obtained through **PCA** and **FA**, both analyses were conducted in parallel using two distinct sets of variables. First, for the **PCA**, *Bartlett's test of sphericity* was performed to verify that the necessary assumptions for this technique were met. Although the initial conditions were satisfied when using all numerical variables, the potential quality of the results remained uncertain. At this stage, we used the *Kaiser–Meyer–Olkin (KMO)* test, which provides a measure of sampling adequacy for each variable to determine their suitability for the analysis. By selecting the most appropriate variables, we defined the first set of variables mentioned: **ESTRATO.x**, **MONTHS**, **WEIGHT**, **HEIGHT**, **MUAC**, and **NUM_FUENT_INGRES**.

First, the **correlation matrix** of the variables is calculated. The *determinant* of the matrix, being greater than 0 but small, indicates a certain degree of multicollinearity among the variables. From the **correlation plots**, it is evident that the variables **MONTHS**, **WEIGHT**, and **HEIGHT** are highly correlated. Conversely, the remaining variables show little to no correlation with each other or with these three.

When **Bartlett's test of sphericity** is applied, the null hypothesis states that the correlation matrix is an *identity matrix*, implying that the variables are uncorrelated. The *chi-squared* (χ^2) value in all cases is high, indicating that the analysis will be significant; with a *p-value* equal to 0, we can reject the null hypothesis. From the **KMO Test**, we obtain an *Overall Measure of Sampling Adequacy (MSA)* of 0.75, which signifies that the analysis is "middling" to "meritorious" (very good), though not perfect.

Once the assumptions were verified, the **prcomp** function was used to generate the **principal components**, the *loadings* (coordinates relative to the components), and the *standard deviations* of each component. By observing the **summary(PCA)** output and applying the **Rule of Abdi et al.**, we retain the first components with *standard deviations* greater than 1 (as the data have been normalized). According to this rule, the study suggests retaining the first two **principal components**. However, considering the **Factor Analysis (FA)** results, if we retain the first three components—matching the number of *latent factors*—we achieve an **explained variance** of 86%.

These plots visually illustrate which variables are best explained by the model: those with longer arrows are well-represented by the corresponding components. For instance, variables such as **ESTRATO.x** and **NUM_FUENT_INGRES** in the DIM1-DIM2 plane exhibit an angle close to 180°, indicating a negative correlation. Conversely, **WEIGHT**, **HEIGHT**, and **MONTHS** display a small angle between them in the DIM1-DIM3 subspace, suggesting a high positive correlation.

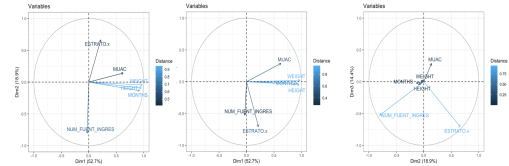


Figure 3: PCA Results

In the DIM1-DIM2 plane, **MUAC** appears associated with this group. Conversely, **NUM_FUENT_INGRES** and **ESTRATO.x** appear close together with a weak representation on the third dimension.

0.2 Factorial Analysis

Although the preliminary study evaluated various sets of variables, the results presented below focus on the second group of analysis, which was identified as the model with the highest explanatory power and internal consistency.

In the initial phase, the Minimum Residuals (*minres*) and Maximum Likelihood (*ml*) models were contrasted to determine which exhibited greater consistency with respect to our data. The *minres* model was selected, as it yielded a higher total explained variance and demonstrated better convergence with the nature of the dataset.

Subsequently, the analysis focused on determining the optimal number of factors to consider in order to achieve the highest total explained variance.

To this end, tests were conducted with different numbers of factors, concluding that a 3-factor model is the most optimal choice, as it explains 75% of the total variance—a result that guarantees high representativeness of the original information.

This diagnosis is supported by the **Scree Plot** analysis and **Parallel Analysis**, which confirm the inflection point at the third factor.

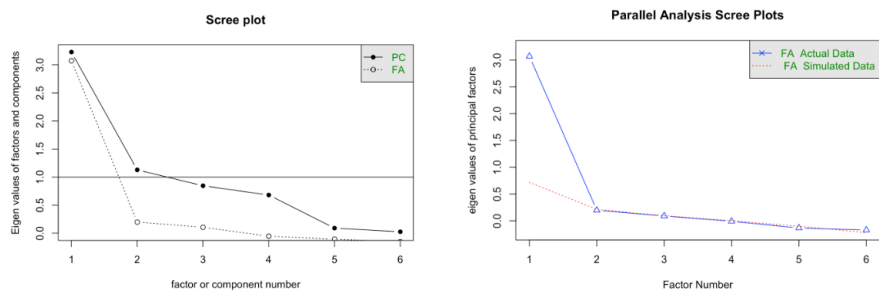


Figure 4: Scree Plot and Parallel Analysis results.

An orthogonal Varimax rotation was applied to maximize the interpretability of the factor loadings. The final solution reveals a clear structure of three latent dimensions.

Factor 1 consolidates integral physical development; **Factor 2** isolates the geographical impact (*stratum*) as an independent dimension; and Factor 3 reveals nuances in the proportionality of the Mid-Upper Arm Circumference (MUAC) in relation to age.

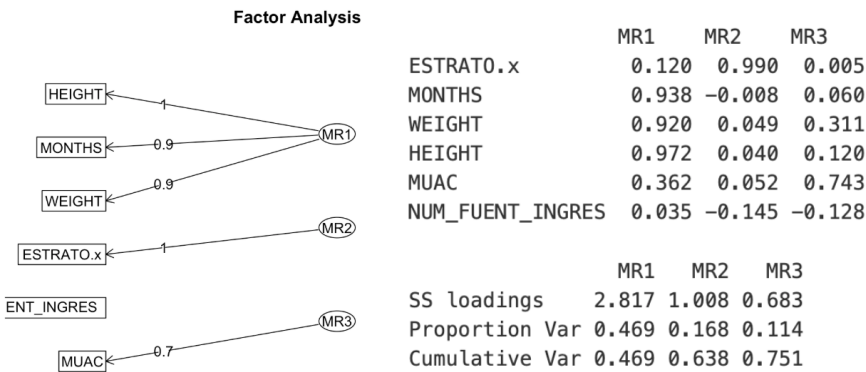


Figure 5: Factorial Analysis Results

0.3 Cluster Analysis

The analysis began with the determination of the optimal number of clusters to ensure the highest representativeness of the data.

The initial dendrogram revealed a primary natural partition within the sample. The height of the upper branches confirmed a significant heterogeneity between two major groups, validating the robustness of this segmentation to isolate distinct profiles.

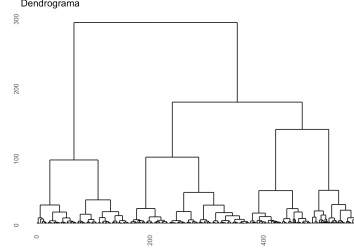


Figure 6: Dendrogram

The selection of a two-cluster solution is definitively supported by the convergence of three fundamental diagnostic criteria:

- **Optimal Cohesion:** The Silhouette Coefficient peaks at $k = 2$, indicating that this configuration yields the highest degree of internal consistency and external separation.
- **Statistical Parsimony:** The Elbow Method (WSS) reveals that the most significant reduction in within-cluster variance occurs at the transition to two groups, identifying the optimal point of diminishing returns.
- **Hierarchical Validation:** The Dendrogram confirms a natural primary cleavage in the sample, with a substantial dissimilarity distance between the two main branches.

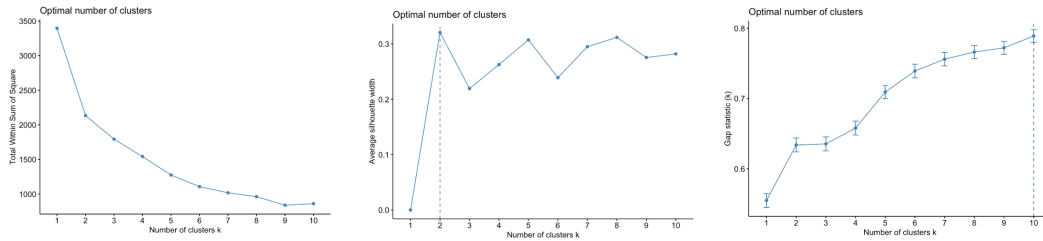


Figure 7: Elbow method, Silhouette Method and Gap Statistical Method

After establishing the two-group framework as the optimal model, we proceeded to characterize the identified subpopulations using the means of the standardized indicators (Z-scores):

Cluster 1: Characterized by negative means across all standardized biometric indicators: MONTHS (-0.85), WEIGHT (-0.83), HEIGHT (-0.83), and MUAC (-0.52). Additionally, it presents a slight deficit in the number of income sources (-0.03) relative to NUM_FUENT_INGRES.

Cluster 2: Characterized by positive values across all indicators: MONTHS (0.79), WEIGHT (0.76), HEIGHT (0.77), and MUAC (0.48), positioning this group above the sample average. It also shows a higher mean in terms of NUM_FUENT_INGRES(0.03).

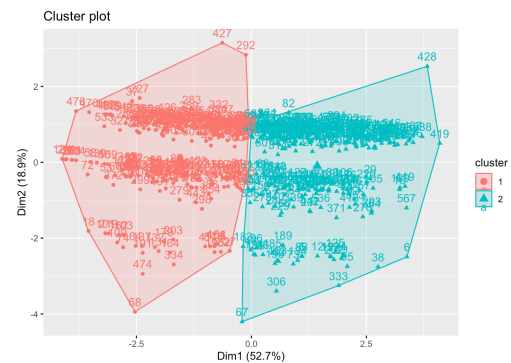


Figure 8: Different partition for the dataset depending on the number of clusters

0.4 Discriminant Analysis

The analysis was conducted using *SUNING* as the grouping criterion. To ensure external validity, the dataset was divided into a training set (80%) and a test set (20%). The assessment of assumptions through graphical representations and the *Shapiro-Wilk test* yielded p-values below 0.05 for most variables, indicating a departure from multivariate normality.

Furthermore, Box's M test resulted in the rejection of the null hypothesis of homogeneity of covariance matrices ($p < 9.84 \times 10^{-16}$). Due to this variance heterogeneity, it was determined that fitting a **Quadratic Discriminant Analysis (QDA)** was the most appropriate approach in this case to generate non-linear decision boundaries, while maintaining **Linear Discriminant Analysis (LDA)** for comparative purposes.

Upon conducting the LDA, descriptive statistics revealed significant differences in anthropometric indicators. The group with stunting (57.32% of the sample) shows an average age of 33.46 months, a weight of 10.91 kg, and a height of 81.73 cm. In contrast, the group with normal growth recorded 30.85 months, 11.98 kg, and 85.61 cm, respectively. The coefficients of the linear discriminant function assigned the highest weights to **WEIGHT** (-0.244) and **HEIGHT** (-0.193), while **STRATUM** (0.01) and **NUM_FUENT_INGRESOS** (0.008) showed the lowest values.

Model validation was performed through a random data partition (80% training, 20% test), achieving a correct classification rate of 82%. The model provided posterior probabilities for observations; for a specific individual test profile, a probability of 97.34% belonging to the healthy group was obtained (LD1 score of -2.62). However, other observations were located in ambiguous zones with probabilities close to the decision threshold (e.g., 0.54 vs. 0.46).

Finally, partition maps for both models represented the classification regions (green for Group 0 and red for Group 1). These plots showed defined boundaries in the cross-tabulations of biometric variables, where the centroids of each group are widely spaced, whereas socioeconomic predictors showed a significant overlap of observations regarding the assigned areas.

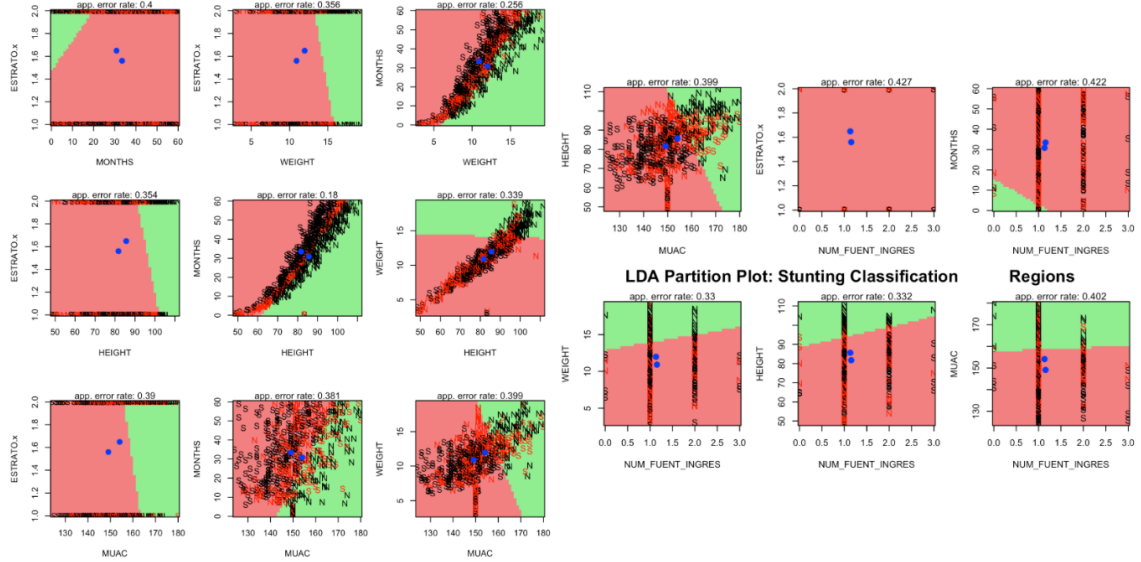


Figure 9: Partition plots

The QDA model reveals significant differences in anthropometric indicators: children in the normal growth category (Group 0) present higher values in Height (85.3 cm vs. 81.3 cm) and Weight (11.89 kg vs. 10.82 kg) compared to the group with chronic malnutrition. These results position biometric variables as the primary drivers of classification, in contrast to factors such as the number of income sources, whose mean similarity (1.14 vs. 1.15) confirms their limited discriminatory utility.

Beyond discrete assignment, the model provides posterior probabilities that allow for an assessment of the reliability of each classification. While some observations are assigned with nearly

100% certainty, others show greater ambiguity when situated near the decision boundary (e.g., probabilities of 0.54 vs. 0.46). This classification structure was validated using confusion matrices and visualized through partition maps. These plots confirm that the classification regions are clearly delimited by biometric variables, where the centroids of each group are more widely spaced, facilitating a more precise detection of nutritional status.

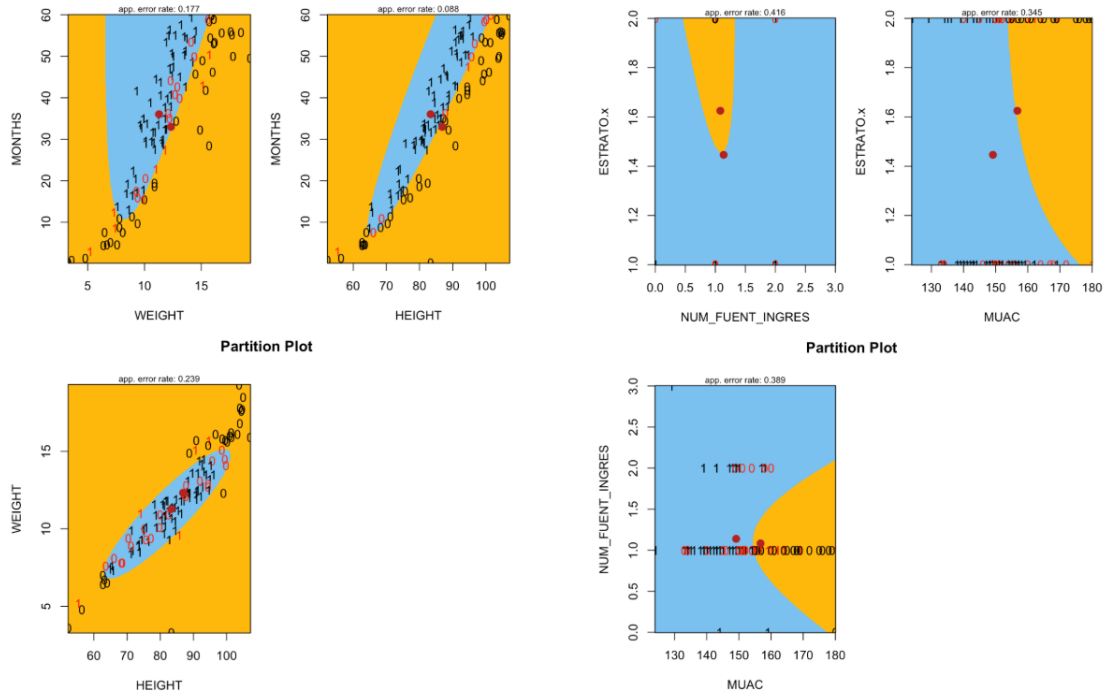


Figure 10: Limits of a linear or quadratic discriminant model for each pair of predictors.

Discussion

0.1 Principal Component Analysis

The initial objective of this technique is to reduce the dimensionality of the studied variables or to be used as a preliminary step for cluster analysis. Based on the results obtained in conjunction with Factor Analysis, and applying the *Rule of Abdi et al.*, we retained the first three principal components, which account for 86% of the explained variance (although retaining only the first two would also yield an acceptable value).

0.2 Factorial Analysis

The central objective of this study is to determine whether the latent dimensions obtained from this analysis remain consistent with the PCA results, thereby ensuring the robustness of our conclusions through multiple approaches.

Through Factor Analysis (FA), the goal of simplifying the data structure was achieved, demonstrating that a three-factor model is the most optimal for explaining 75% of the variability. This model effectively separates physical development from geographic and nutritional components.

In conclusion, FA has allowed for the reduction of information into three actionable axes, proving that to understand malnutrition in the region, it is essential to distinguish chronic growth from territorial determinants.

0.3 Cluster Analysis

The primary objective of this study was to identify and characterize nutritional vulnerability profiles in the child population using multivariate segmentation techniques.

The outcome of this goal is evidenced in the identification of Cluster 1, which groups individuals with a clear biological risk profile, showing systematic deficiencies in weight, height, and mid-upper arm circumference (MUAC) compared to the stability observed in Cluster 2.

In conclusion, the analysis demonstrates that malnutrition in the region is not uniform, allowing future interventions to be focused on the identified profile with the greatest biometric precariousness.

0.4 Discriminant Analysis

The primary objective of this analysis was to evaluate the capacity of biometric and socioeconomic indicators to discriminate and predict chronic malnutrition (SUNTING) in children. The results confirm that this objective was successfully achieved through Discriminant Analysis.

The model demonstrates that anthropometric variables—specifically height and weight—are the most powerful predictors, showing clear decision boundaries and high discriminatory weight. Conversely, the negligible impact of factors such as stratum and income sources suggests that, in this sample, clinical metrics are more immediate indicators of nutritional status than household economic variables.

Despite the non-fulfillment of multivariate normality, the 82% accuracy rate validates the robustness of the LDA and QDA models. This confirms that SUNTING can be effectively categorized through a reduced set of biometric markers, providing a reliable tool for focused nutritional interventions.

Conclusion

Our analysis of children under five years of age in Guatemala's "Corredor Seco" began with a review of the study provided alongside the survey. This study proposes recommendations such as establishing emergency funds to cover, for instance, malnutrition treatments. However, how can we determine when these funds are truly necessary?

Through our study, we have sought to analyze all available variables to identify those that perform best in two ways: those that most accurately represent childhood malnutrition and those that are the most accessible to obtain. This led us to focus primarily on physical indicators: MONTHS, WEIGHT, and HEIGHT; a geographic variable: ESTRATO.x; and a socioeconomic variable: NUM_FUENT_INGRES.

Finally, we selected the health variable MUAC, as the study emphasized the importance of introducing mid-upper arm circumference measurement in health centers.

By means of **Principal Component and Factor Analysis**, the structure of the selected subgroup was successfully simplified from seven variables to three groups that explained nearly 80% of the variance: child physical development, the geographic component, and nutritional balance.

Via **Cluster Analysis**, the existence of two distinct risk profiles was identified: the first represents the most vulnerable population (characterized by systematic deficiencies in weight, height, and mid-upper arm circumference), while the second shows greater biometric and financial stability.

Lastly, the **Discriminant Analysis** demonstrated that, although location and economy are factors that define the household context, the strongest predictors for identifying chronic malnutrition (SUNTING) are strictly related to human body measurements.

In short, with an accuracy of 82%, it is confirmed that height and weight are the critical drivers of classification, although structural geographic and economic deficiencies can be addressed simultaneously.